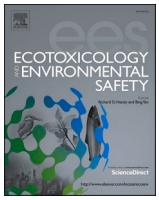
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Hepatotoxicity assessment investigations on PFASs targeting L-FABP using binding affinity data and machine learning-based QSAR model

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ABSTRACT

Per- and polyfluoroalkyl substances (PFASs) are persistent organic pollutants that have been detected in various environmental media and human serum, but their safety assessment remains challenging. PFASs may accumulate in liver tissues and cause hepatotoxicity by binding to liver fatty acid binding protein (L-FABP). Therefore, evaluating the binding affinity of PFASs to L-FABP is crucial in assessing the potential hepatotoxic effects. In this study, two binding sites of L-FABP were evaluated, results suggested that the outer site possessed high affinity to polyfluoroalkyl sulfates and the inner site preferred perfluoroalkyl sulfonamides, overall, the inner site of L-FABP was more sensitive to PFASs. The binding affinity data of PFASs to L-FABP were used as training set to develop a machine learning model-based quantitative structure-activity relationship (QSAR) for efficient prediction of potentially hazardous PFASs. Further Bayesian Kernel Machine Regression (BKMR) model disclosed flexibility as the determinant molecular property on PFASs-induced hepatotoxicity. It can influence affinity of PFASs to target protein through affecting binding conformations directly (individual effect) as well as integrating with other molecular properties (joint effect). Our present work provided more understanding on hepatotoxicity of PFASs, which could be significative in hepatotoxicity gradation, administration guidance, and safer alternatives development of PFASs.

1. Introduction

Per- and polyfluoroalkyl substances (PFASs) have been widely applied in numerous industrial and commercial products since 1950s (Amstutz et al., 2022; Gao et al., 2020; Presentato et al., 2020), which has led to the increase of distributions in various environmental media and biota samples (Dai et al., 2006; Penland et al., 2020; Han et al., 2023). Of particular concern are perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS), due to the extremely widespread application and distribution in environment (Zhu et al., 2022). Although the Stockholm Convention in 2009 has issued restrictions on the production and usage of PFOA, PFOS and related compounds, many alternatives have still appeared on the market (Presentato et al., 2020; Shan et al., 2021). At present, about 5000 kinds of identified PFASs have been registered and more PFASs are still unknown, which possess persistence and bioaccumulation characteristics in human body (Feng et al., 2019; Liu et al., 2019; Dickman and Aga, 2022). Therefore, accurate prediction and assessment of the toxicity of PFASs to humans are essential for long-term development goals in the field of environmental health.

It has been reported that PFASs have toxic effects on reproduction, immune and endocrine function (Sunderland et al., 2019). For instance, PFOS and 6:2 Cl-PFESA can regulate the expression level of key mediators in phosphatidylinositol metabolism pathways and enzyme in glycerophosphoserines biosynthesis to cause hepatotoxicity (Li et al., 2023). PFASs can also reduce the PC/PE ratio of liver cell membrane, inhibiting the secretion of VLDL and promoting the accumulation of triglyceride in hepatocytes (Li et al., 2023). The interaction between PFASs and cytochrome P450 has been proposed as a potential key molecular event of leading to PFAS-induced hepatotoxicity (Amstutz et al., 2022). Besides, PFOS can cause hepatic steatosis and pathological micro/macro vacuolation with a superfluous triglyceride level occurred in zebrafish (Yi et al., 2019). These studies suggested that liver is the

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significant target organ of PFASs. The accumulation of PFASs in liver may result in and even exacerbate the hepatotoxicity (Huang et al., 2022; Chen et al., 2021). Recent study indicated that liver weight, hepatosomatic index, triglyceride content and cholesterol content were positively correlated with the liver PFAS concentration in *Rana nigromaculata* (Lin et al., 2022). It has also demonstrated a strong positive correlations between the –log dissociation constants (–logKd) of PFASs for liver fatty acid binding protein (L-FABP) and accumulation of PFASs in liver (Cao et al., 2019; Jia et al., 2022). Therefore, binding affinity to L-FABP is a major contributing factor to the potential bioaccumulation of PFASs in liver tissues. It has been verified that several typical PFASs, such as PFOA, PFOS and related perfluorooctanesulfonamide-based fluorochemicals (PFOSAs) can bind to L-FABP due to the structural similarity to its natural substrate (fatty acids) (Luebker et al., 2002; Sheng et al., 2016). The binding to L-FABP facilitates the transport of PFASs into hepatocytes, thereby initiating some relevant hepatotoxic effects. Taken together, L-FABP is a pivotal protein that affects hepatotoxicity due to the accumulation of PFASs in the liver. Regarding to the critical role of L-FABP in the hepatotoxicity of PFASs, a growing number of studies aiming to assess the toxicity of PFASs according to L-FABP binding affinities have been proposed (Cao et al., 2019; Cheng and Ng, 2018). Currently, most toxicity data available merely refer to a handful of chemicals, such as PFOA and PFOS, while hepatotoxicity data of most PFASs are notably deficient and, in the case of emerging replacements and by products, the data are completely unexplored (Fenton et al., 2021). Therefore, elucidating the structure-based interactions between PFASs and L-FABP is essential to predict the hepatotoxicity of novel PFASs.

Although only a limited number of studies have investigated the binding affinities of PFASs to L-FABP (Gao et al., 2019; Sheng et al., 2016), a recent nontarget screening method using Size-Exclusion Chromatography-Coelution (SECC) and nontarget analysis (NTA) has identified that PFASs were with high binding affinity to L-FABP (Yang et al., 2020). And the quantitative relationship between PFASs and L-FABP binding has been disclosed through experimental exposure and QSAR model combined with molecular dynamics (MD) simulations (Cao et al., 2019), which provided an elaborate computational model to predict the L-FABP binding affinities of some specific PFASs. But given the sheer number and structural diversity, a larger quantity of PFASs with broader scope of categories should be investigated. Notably, in silico prediction approaches, such as quantitative structure-activity relationship (QSAR) models, hold promise for the quantitative assessment of chemical risk, particularly when the training set is of high quality. These methods have already been utilized to evaluate the endocrine disrupting effects of PFASs (Azhagiya Singam et al., 2020; Lin and Chou, 2022; Yu et al., 2022; Lavado et al., 2022). Therefore, the use of in silico approaches to predict the hepatotoxicity of PFASs towards L-FABP is of significant interest. Given the diversity of PFASs compound properties, considering the impact of various molecular characteristics of PFASs on their affinity to L-FABP is of particularly important. However, the use of multivariate parametric regression methods may be inadequate in addressing collinearity and non-additivity issues. Previous studies have primarily focused on investigating the influence of single molecular features, which may be inadequate regarding to understanding the real impact of multiple molecular features. Fortunately, the Bayesian Kernel Machine Regression (BKMR) models have recently been introduced as a novel approach capable of assessing the combined effects of molecular features, which could provide insight into the hepatotoxicity of PFASs comprehensively (Bobb et al., 2015; Roostaei et al., 2021).

In the present study, we calculated the binding affinities of 1599 PFASs to L-FABP and elucidated the dominant binding site of target protein in PFASs-induced hepatotoxicity. By using the binding affinity data as training set, a machine learning-based quantitative structure-activity relationship (QSAR) model was subsequently developed for the efficient prediction of potentially hazardous PFASs. Further individual and joint impacts of confounding factors investigation with nonparametric statistical approach (BKMR) revealed that molecular flexibility was the determinant of PFASs-induced hepatotoxicity. The present work could be significative in hepatotoxicity gradation, administration guidance, and safer alternatives development of PFASs.

2. Materials and methods

2.1. Model preparation

L-FABP is of particular interest as it can bind two fatty acid molecules simultaneously. Both two ligand binding sites need to be studied in the present work and all the resolved structures of L-FABP are highly identical. The ligand-receptor complex that contains two ligands in the two binding sites would be more convenient for the present study, which can distinctly indicate the binding site of PFAS and provide more accurate docking sites for molecular docking process. Therefore, the X-ray crystal structure of human liver-fatty acid binding protein complexed with two molecules of oleic acid recorded in Protein Data Bank (https://www.rcsb.org/, PDB code: 2LKK) was utilized to construct the initial receptor model. The protein and natural ligand (oleic acid) parts were extracted and processed for subsequent molecular docking experiments. A ligand database comprising 1599 compounds (listed in Supplemental Material_1) was constructed based on a recent study that reported the discovery and characterization of PFASs in environmental and human samples (Liu et al., 2019). All structures in the database were optimized using the OPLS4 force field, and the protonated states of ionizable groups were defined at pH 7.0 ± 0.2 (Kontoyianni, 2017), which simulated the slightly fluctuating pH conditions in the physiological liver environment.

2.2. Molecular fingerprint calculation

A total of 13 molecular fingerprints and 192 categories of 2D molecular descriptors were utilized in this study, generated from an integrated web-based platform for molecular descriptor and fingerprint computation (http://www.scbdd.com/chemdes/). The molecular fingerprints included CDK fingerprint (1024 bits), CDK extended fingerprint (1024 bits), Graphonly fingerprint (1024 bits), AtomPairs2D fingerprint (780 bits), AtomPairs2D count fingerprint (780 bits), E-state fingerprint (79 bits), Klekota-Roth fingerprint (4860 bits), Klekota-Roth Count fingerprint (4860 bits), PubChem fingerprint (881 bits), Substructure fingerprint (307 bits), Substructure Count fingerprint (307 bits), MACCS fingerprint (166 bits) and Morgan fingerprint (1024 bits). Detailed explanations on these molecular fingerprints were provided in Table S1. Additionally, 192 categories of 2D molecular descriptors were calculated by MOE software (version 2014.09). Prior to machine learning analysis, all molecular fingerprints and descriptors were standardized as follows:

$$X_i = (X_0 - \mu) / \sigma$$

where X_i is the standardized value of fingerprints or descriptors, X_0 is the original value, μ is the feature mean, and σ is the feature standard deviation.

2.3. Molecular docking

Molecular docking was performed by using the Maestro module in Schrödinger software package. Protonated states of titratable residues in receptor were calculated at the same pH (7.0 ± 0.2) as the ligand preparation (Kontoyianni, 2017). The receptor was relaxed with the OPLS4 force field at 310 K. Glide docking procedure was employed for the current docking study, with centroid of oleic acid defined as the center of the docking grid. The size of the grid was set to 25 * 25 * 25 $Å^3$. Finally, flexible molecular docking based on induced fit theory was

executed, and results (binding mode and docking score) with the best docking score were recorded.

2.4. Correlation analysis

In order to assess the credibility of molecular docking, a regression analysis was conducted using IBM SPSS Statistics 25. Dissociation constants (K_d) of PFASs towards L-FABP from previous studies (Yang et al., 2020) and the binding energy obtained from molecular docking were used as the independent and dependent variables, respectively. The Pearson Correlation Coefficient (r) was selected to evaluate the degree of explanation of independent variable towards dependent variable, namely K_d and binding energy.

2.5. Machine learning

Firstly, missing values were filled with 0 and variance filtering (variance is 0) was applied (Lin et al., 2021). Next, Scikit-learn, based on the Python language, was utilized for constructing the predictive model. Prior to training, the data sets including 829 components for non-portal site and 933 components for portal site were separately divided into training and verification sets at a ratio of 4:1 at random. Two categories with five machine learning algorithms, including classical and popular regression algorithm (Ridge and Lasso), and other algorithms transformed from classification algorithms (Decision Tree (DT), Support Vector Machine (SVM) and Random Forest (RF)), were employed for constructing the models. The hyperparameters of each algorithm were optimized using the grid search and learning curve methods. The best hyperparameter values were summarized in Table S2 and detailed parameter values were provided in Supplementary Materials_2. The predictive ability and robustness of these regression models were evaluated using ten-fold cross-validation (internal validation) and external validation (Du et al., 2017). The coefficient of determination (R^2) was used to assess the regression performance of the results obtained from the machine learning algorithms.

2.6. BKMR

Although ridge regression and other statistical methods have solved the multicollinearity problem, they are incapable of modeling the nonlinear, non-additive, or high dimensional functional relationships. By using the kernel machine regression model, BKMR can provide in-depth analysis on the potentially complex nonlinear relationship or/and interaction. It can fit various effects produced by molecular characteristics, such as relative contributions on the overall effect, joint effects of multiple molecular characteristics, and bivariate exposure-response functions of molecular features. In this study, R version 4.2.1 (R Foundation for Statistical Computing, Vienna, Austria) was employed to implement BKMR, which was established as follows (Bobb et al., 2015):

$$Y_i = h(x_i) + z_i^{T} \beta + e_i$$

where Y_i is the dependent variable or response variable, which represents hepatotoxicity, namely binding affinity of PFASs to L-FABP, x_i represents molecular characteristics (such as Pol, nA, Hyd, nC, nF, Brs, Bs, Flex, logP, Varea, Vvol, Wei), $h()$ is a Gaussian kernel exposure-response function, z_i is the covariate of x_i, containing a set of potential confounders (extrinsic affecting factors distinguished from the intrinsic molecular characteristics of PFASs, and no potential confounder was included in the present study) (Bobb et al., 2015), T is the transpose operator, β is the linear regression coefficient associated with the input variable x_i, and e_i refers to the residuals.

Posterior inclusion probabilities (PIPs) of molecular descriptors were calculated via a hierarchical variable selection method (Bobb et al., 2015) with 25,000 iterations using a Markov chain Monte Carlo sampler, to determine the contribution of each descriptor. The relationship between a specific characteristic and affinity was calculated while controlling for the other 11 PFASs characteristics at the median. Linear or nonlinear relationships could be directly observed as a result. Additionally, the change in binding force was calculated at the median, while controlling for all PFASs characteristics at specific percentiles (25th, 50th, 75th), when the single PFASs feature increased by an inter-quartile range (IQR). Moreover, controlling for the other 10 PFASs characteristics at the median, the relationship between one characteristic at the 10th, 50th, 90th percentiles and another characteristic elevating an IQR with affinity was calculated.

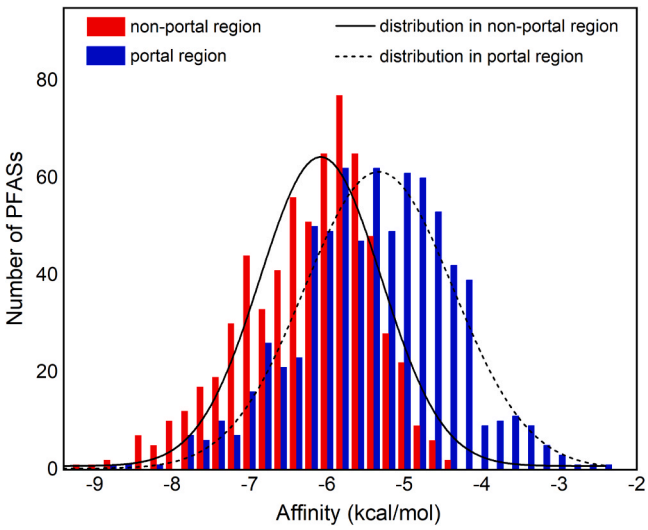


Fig. 1. Distribution on affinity of PFASs binding to portal and non-portal region of L-FABP.

3. Results and discussion

3.1. Molecular docking and accuracy verification

Due to the limited availability of toxicity/affinity data towards L-FABP for the huge number of PFASs, molecular docking was employed to generate sufficient binding affinity data in the present study. L-FABP is comprised of a β-barrel composed of 11 anti-parallel β-sheets with two α-helices closing one end. Unlike other intracellular lipid-binding proteins, L-FABP is unique in containing two substrate binding sites, namely portal and non-portal region (Fig. S1) (He et al., 2007; Sharma and Sharma, 2011). While previous studies mainly focused on the interaction between PFASs and the non-portal region of L-FABP (Yang et al., 2020), the portal region, as a substrate binding site, also warrants detailed investigation. Therefore, the binding of PFASs towards both the portal and non-portal regions of L-FABP was analyzed using molecular docking. As a result, 766 PFASs and 653 PFASs were successfully docked in the portal and non-portal regions, respectively, with a total of 621 PFASs exhibiting considerable binding affinity towards both two regions.

For toxic compounds, the tighter binding to biological target generally exhibits the stronger toxicity (Cheng et al., 2021). To verify the correlation between binding affinity to L-FABP and actual hepatotoxicity of PFASs, the correlation between binding energy from in silico molecular docking and dissociation constant from in vitro fluorescence displacement experiments was analyzed. The binding affinity data of PFASs to L-FABP were generated with the higher value to the two pockets, as a result, all the affinity data were derived from non-portal region. The Pearson correlation coefficient between dissociation constant and binding affinity was 0.79 (Fig. S2), indicating that the docking score of PFASs binding to L-FABP generated by molecular docking can

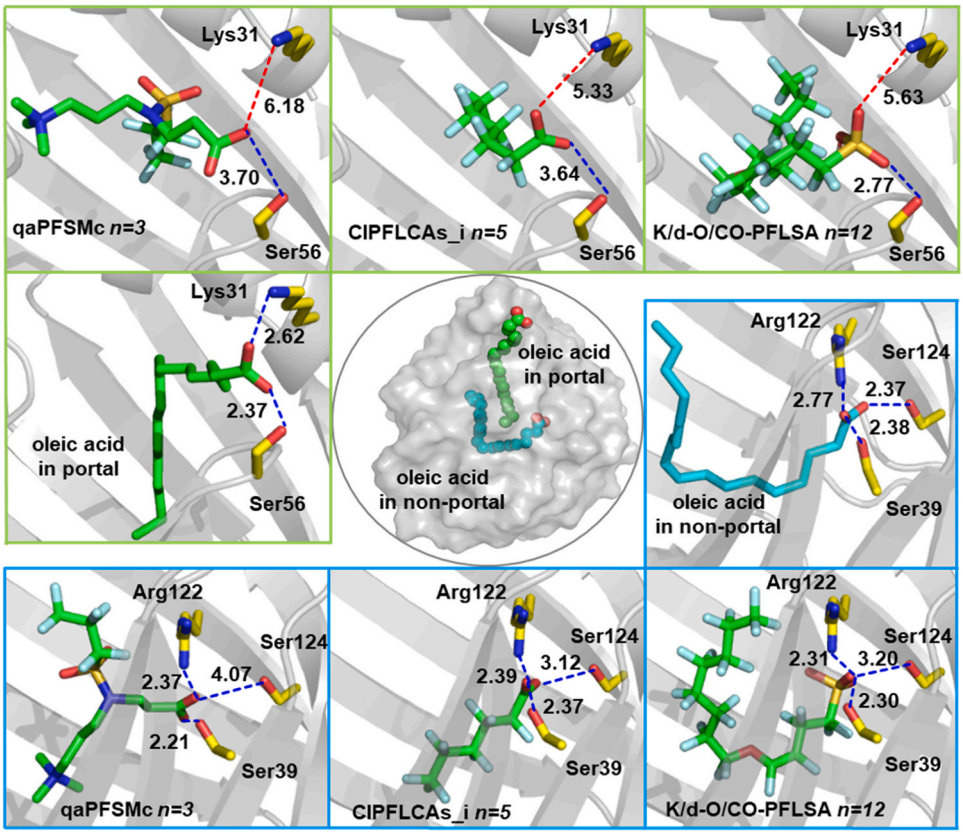


Fig. 2. The binding modes of three typical PFASs and oleic acid to the portal region (green border) and non-portal region (blue border) of L-FABP. Hydrogen bond or salt bridge interactions are labeled with blue dotted line, red dotted line represents no obvious interactions. Distances are given in angstrom.

reflect the potential hepatotoxicity of PFASs. As shown in Fig. 1, affinities of PFASs binding to the portal site ranged from −8.92 to −2.44 kcal/mol with a mean of −6.33 kcal/mol, and from −9.42 to −4.56 kcal/mol with a mean of −5.38 kcal/mol to the non-portal site. The natural ligand exhibits a significantly lower binding affinity compared to PFASs at either of the two sites (−4.34 kcal/mol in portal and −3.46 kcal/mol in non-portal region), demonstrating the potent competitor role of PFASs. PFASs showed a higher affinity for the non-portal site than the portal site ($p < 0.01$). The difference of about 1 kcal/mol on the median for affinity of PFASs to the two sites corresponds to about six-fold change in activity (IC_{50}), revealing the biologically significant difference on inhibitory activity. Moreover, the non-portal site of L-FABP is occupied preferentially when binding with natural substrate or competitive inhibitors. In conclusion, the non-portal pocket of L-FABP, the prioritized binding site of PFASs, is more sensitive to the production of PFASs toxicity.

3.2. Binding mode analysis

For the natural state of L-FABP, two oleic acid molecules are combined in the portal and non-portal substrate binding sites. As displayed in Fig. 2, hydrogen bond or salt bridge interactions are formed between

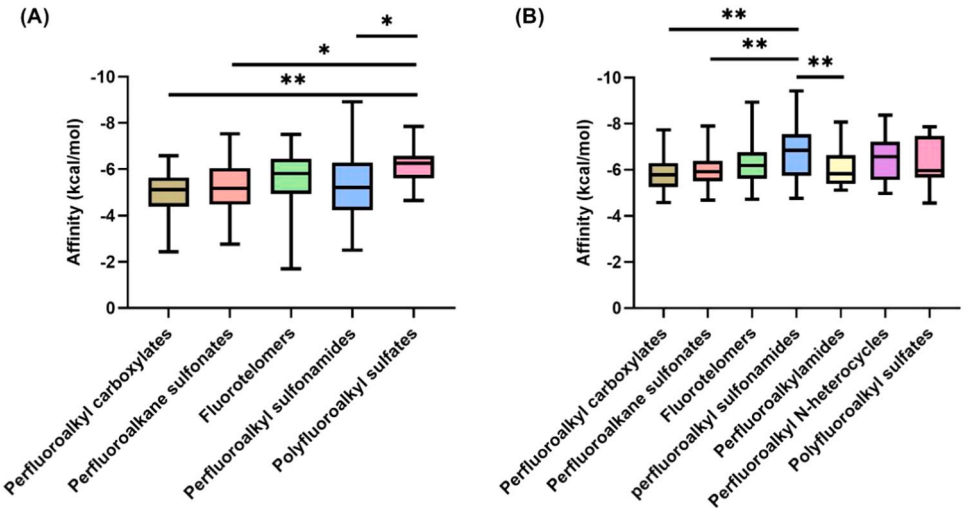


Fig. 3. Comparison on binding affinities of different types of PFASs to L-FABP in the portal region (A) and non-portal region (B). * $p < 0.05$ and * * $p < 0.01$.

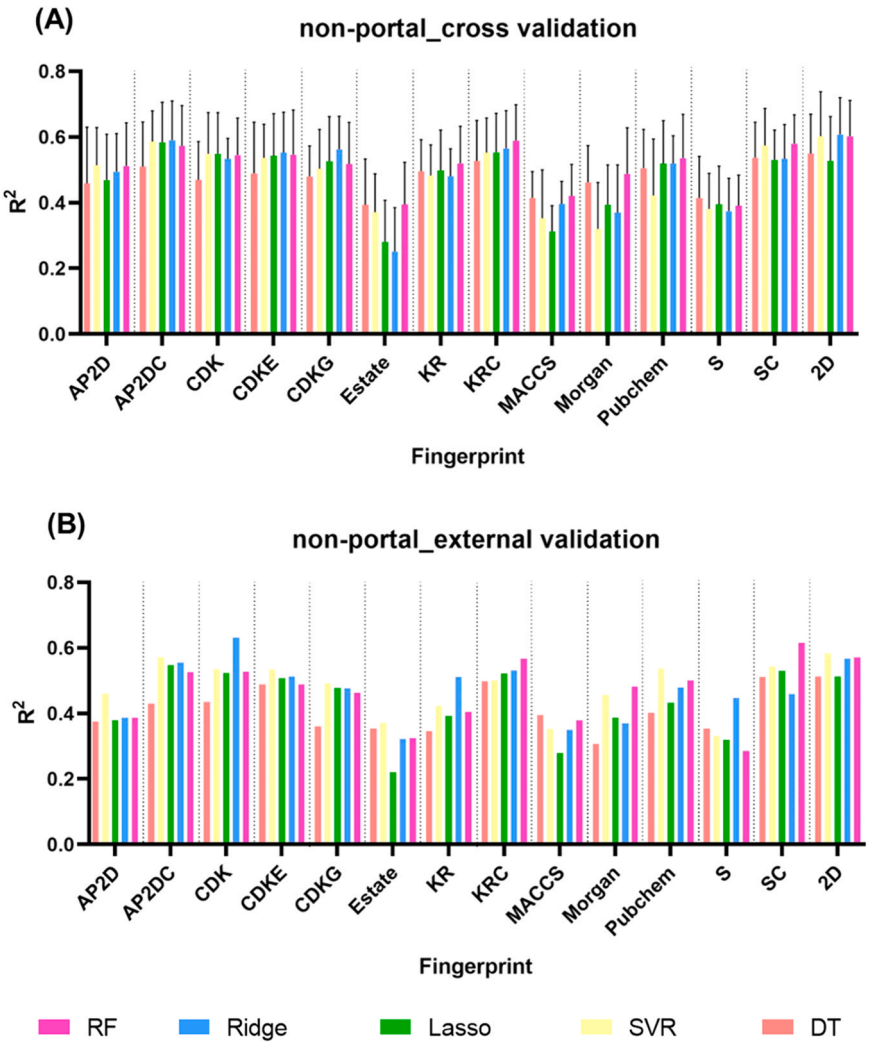


Fig. 4. Performance of toxicity prediction models of PFASs towards L-FABP assessed by cross-validation and external validation. Abbreviations: AP2D, AtomPairs2D; AP2DC, AtomPairs2D Count; CDKE, CDK Extended; CDKG, CDK Graphonly; KR, KlekotaRoth; KRC, KlekotaRoth Count; MACCS, MACCSkeys; S, Substructure; SC, Substructure Count; 2D, 2D descriptor; RF, Random Forest; SVR, support vector regression; DT, Decision Tree.

oleic acid and polar residues, such as Lys31 and Ser56 in the portal region, and Arg122, Ser39, and Ser124 in the non-portal region. Three typical PFASs that refers to short chain (ClPFLCAs_i), long chain (K/d-O/CO-PFLSA) and branched chain (qaPFSMc) were selected for the binding mode analysis (Fig. 2). The results indicate that PFASs can bind to the non-portal region with the same binding mode as oleic acid, demonstrating a tight binding to L-FABP. However, in the portal region, only Ser56 contributes slightly to the binding, and no obvious interaction is found with Lys31 (the distances are greater than 5 Å shown in Fig. 2). The different binding modes in the two sites could be a reasonable explanation to the different binding affinity of PFASs to portal and non-portal region of L-FABP.

PFASs can also be categorized by the acid groups, and the binding affinity for different types of PFASs to L-FABP was summarized in Fig. 3. Categories with a low success rate or small quantity in docking analysis were excluded, resulting in five categories of PFASs for portal region affinity analysis and seven categories for non-portal region affinity analysis. The medians of the considered categories of PFASs were −6∼−5 kcal/mol in portal region and −7∼−6 kcal/mol in non-portal region, revealing the high affinity of PFASs to the inner site of L-FABP again. Polyfluoroalkyl sulfates exhibited the strongest binding affinity to portal region of L-FABP ($p < 0.05$), and perfluoroalkyl sulfonamides provided the strongest binding to non-portal region ($p < 0.01$).

3.3. Development and characterization of prediction models

The non-portal region has been identified as the prioritized binding site of PFASs, so the binding affinity data with non-portal region of L-FABP were adopted for training and validation set of the subsequent hepatotoxicity prediction model of PFASs. A total of 70 regression models, comprising five machine learning algorithms and fourteen types of molecular descriptors, were established for the machine learning model. Performances of the established models were evaluated through cross-validation (internal validation) and external validation. As depicted in Fig. 4, the cross-validation values ranged from 0.25 to 0.61, while the external validation values ranged from 0.22 to 0.63. Previous reported prediction accuracies for chemicals with limited toxicity were ranged from 0.30 to 0.55 (Lawal et al., 2020; Singh et al., 2022; Tashchilova et al., 2022), and the accuracies of the present prediction models were higher than the previous reported virtual screening strategy (Li et al., 2013). Therefore, a satisfactory model based on the non-portal site of L-FABP was successfully developed to predict the hepatotoxicity of PFASs. In terms of prediction performances, 2D molecular descriptors-Ridge algorithm model with the highest cross validation score (0.61) and competitive external validation score (0.57) was considered as the optimal prediction model to estimate the L-FABP toxicity of PFASs.

The top ten models based on performance of cross-validation were

Table 1
PIPs values of PAFSs molecular descriptors when binding with L-FABP.

Molecular Descriptors	PIPs
Flex	**1.00**
nC	**1.00**
Pol	**1.00**
Varea	0.92
Bs	0.85
nA	0.72
Vvol	0.69
Hyd	0.59
Brs	0.58
logP	0.56
Wei	0.51
nF	0.43

Abbreviations: PIPs, posterior inclusion probability; Flex, molecular flexibility; nC, number of carbon atoms; Pol, sum of atomic polarizabilities; Varea, Van der Waals surface area; Bs, number of single bonds; nA, number of atoms; Vvol, Van der Waals volume; Hyd, number of hydrophobic atoms; Brs, number of rotatable single bonds; logP, log octanol/water partition coefficient; Wei, molecular weight; nF, number of fluorine atoms.

also provided (Fig. S3). The cross-validation scores ranged from 0.57 to 0.61, and external validation ranged from 0.53 to 0.62. The performances of different types of structural fingerprints in predicting PFASs toxicity towards L-FABP were evaluated, and 2D molecular descriptors and AtomPairs2D Count fingerprint were found to have better performance capabilities than other chemical descriptors. These two descriptors showed higher average metric values of cross-validation scores (0.60–0.61 for 2D molecular descriptors and 0.57–0.59 for AtomPairs2D Count fingerprint). Moreover, six out of the top ten models were developed based on 2D molecular descriptors and AtomPairs2D Count fingerprint (three for 2D molecular descriptors and four for AtomPairs2D Count fingerprint). Therefore, the results suggested that 2D molecular descriptors and AtomPairs2D Count fingerprint were good recommendations as chemical descriptors for the development of the prediction models of PFASs toxicity towards L-FABP.

Furthermore, the performance of different machine learning algorithms in hepatotoxicity prediction was analyzed. The top ten models covered four algorithms, among which Ridge had the highest average metric values of cross-validation (0.60–0.61) and Random Forest had the highest frequency of occurrence. The results indicated that Ridge and Random Forest could achieve better results than other three machine learning methods. Unlike the regression algorithms transformed from the classification algorithms (Regression Tree, Support Vector Regression, etc.), Ridge regression is a classical and popular regression algorithm owing to its simplicity, computational expedience, and its robustness in different data regimes (Wan et al., 2014). Regularization term was employed by this classical regression algorithm to form a penalty on the original loss function to deal with the multicollinearity problem. In a certain range of changes, eliminating multicollinearity may improve the generalization ability of the model to a certain extent. Random Forest is one of the most widely used algorithms with excellent performance, speed, and generalizability (Carracedo-Reboredo et al., 2021), and it has proven to be practical in the construction of the prediction models of PFASs hepatotoxicity towards L-FABP.

In addition, the binding affinity data with portal region of L-FABP were also used to train machine learning models with the same regression models. Results showed that the performances of portal models were particularly poor with the best validation score of only 0.30 for cross-validation and 0.35 for external validation (Fig. S4). The unsatisfactory performance on machine learning model for portal region may be attributed to the higher flexibility and solvent accessibility of the binding site, which makes it difficult to correlate PFASs molecular features with binding affinity and further hinders the training of machine learning model.

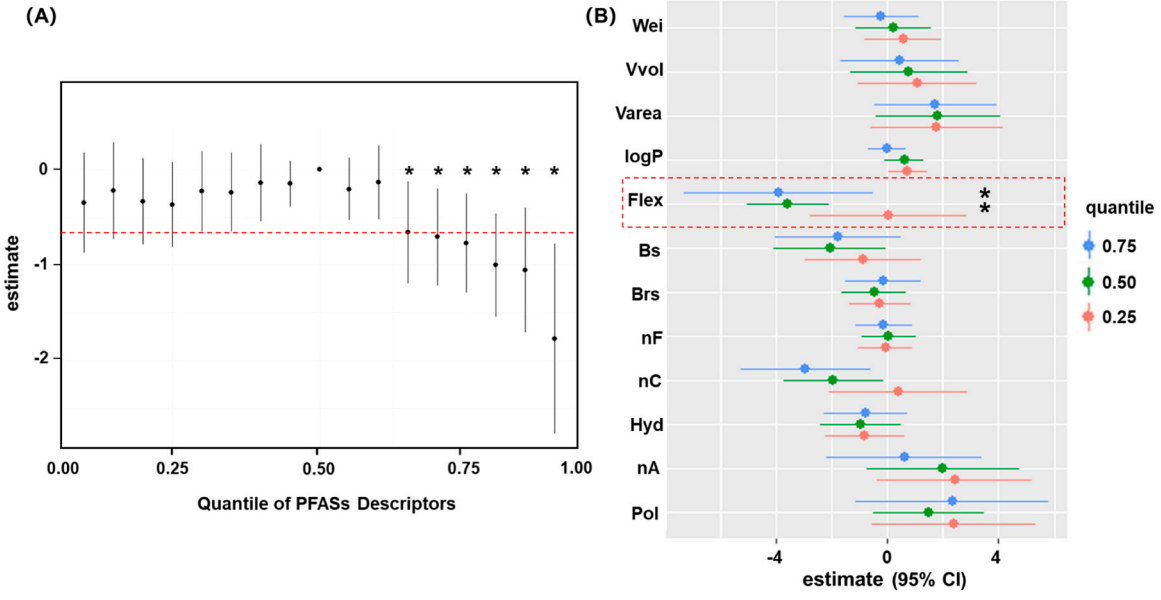


Fig. 5. . (A) Joint effects of exposure to a mixture of PFASs Descriptors on the affinity levels to L-FABP in BKMR models. Results were obtained by comparing the effects of exposure to 12 PFASs descriptors simultaneously at various percentiles (10th to 90th) with the effects of exposure fixed at their 50th percentile. "Estimate" represents the level of affinity with all molecular features at a specific percentile (P_{10}-P_{90}). Effect estimates are represented by circles, while 95 % confidence intervals are depicted by black vertical lines. A red horizontal dotted line is used to represent −0.65. *$p < 0.05$. (B) Individual effects of PFASs descriptors on the affinity levels in BKMR models. The data were plotted by comparing the affinity levels caused by individual descriptors from their 25th to 75th percentile with all other descriptors held at the corresponding quantiles (25th, 50th, or 75th percentile). "Estimate" represents the level of affinity that the molecular characteristic increases an IQR when others are fixed at a particular percentile (P_{25}, P_{50}, P_{75}). Effect estimates are represented by circles, while colored horizontal lines depict 95 % CIs. *$p < 0.05$.

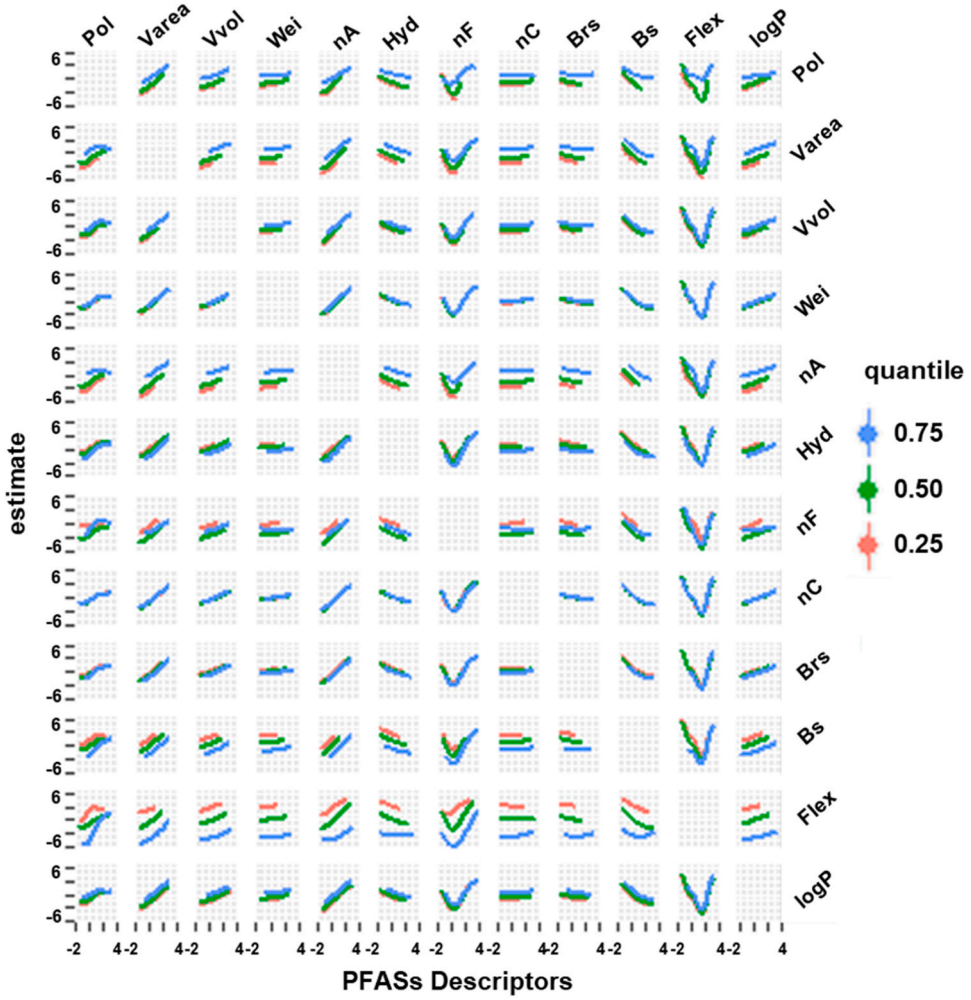


Fig. 6. Bivariate exposure-response functions for PFASs descriptors. "Estimate" represents the level of affinity when the descriptor on the upper coordinate axis holding at 25th quantiles and the other descriptors on the right longitudinal axis held at different quantiles (25th, 50th, and 75th percentiles) with the rest 10 descriptors held at the median. Data were estimated using the Bayesian Kernel Machine Regression.

3.4. Joint/individual effects of different PFASs descriptors on binding ability of PFASs and L-FABP

BKMR was employed to evaluate the contributions of molecular characteristics on binding ability to L-FABP, then uncovering the structure-function relationship of PFAS to hepatotoxicity. By determining the significance of descriptors through PIPs, it showed that molecular flexibility (Flex), the molecule polarization (pol) and the number of carbon atoms (nC) were the most important factors to affecting the affinity to L-FABP (Table 1, PIPs=1.00). Further univariate effect relationship analysis indicated that all the three molecular descriptors were non-linearly correlated to the binding affinity (Fig. S5). The flexibility of PFASs generally increases with the number of carbon atoms, therefore, the two descriptors, nC and Flex, exhibited a similar influence tendency on binding affinity. High flexibility or more carbon atoms can provide more conformations, which is favorable for binding to the target. However, the affinity would decrease as PFASs could no longer enter the protein pockets if the number of carbon atoms increases too much. Besides, the descriptor of molecule polarization was found with irregular influence on binding affinity.

To investigate the overall impact of molecular characteristics on binding affinity between PFASs and L-FABP, joint effects analysis was performed by using BKMR models. As displayed in Fig. 5A, positive overall association with statistical significance between molecular characteristics and binding affinity emerged when all descriptors were above the 65th percentile, indicating that the joint action of various molecular characteristics could increase the affinity to L-FABP. To further identify the driving factors in the overall association between characteristics and affinity, single factor effects analysis was conducted by assessing the change in affinity linked with a change in individual descriptors from the 25th to 75th percentile while maintaining other descriptors at varying quantiles (25th, 50th, or 75th percentile) (Fig. 5B). Results showed that a change in molecular flexibility level from the 25th to 75th percentile was associated with a decrease of 0.03 (95 % CI: −2.78, 2.84), −3.58 (95 % CI: −5.03, −2.12), and −3.90 (95 % CI: −7.28, −0.52) in affinity levels, indicating that flexibility is the determinant of hepatotoxicity caused by PFASs. Further bivariate effect function analysis showed that the three functional variation curves of polarizability intersected as the flexibility level increased from the 25th percentile to the 75th percentile (Fig. 6), indicating the significant interaction of molecular flexibility on polarizability. In conclusion, flexibility was identified as the determinant in PFASs-induced hepatotoxicity, it can not only directly affect binding conformations with individual effect but also integrate on other molecular properties with joint effect to influence the binding of PFASs to L-FABP.

4. Conclusions

In the present study, a series of in-silico approaches were employed to investigate the binding of PFASs to L-FABP, which could be an

effective evaluation of PFASs-induced hepatotoxicity. By using molecular docking, the inner site of L-FABP was identified as the prioritized binding site of PFASs. PIPs analysis disclosed that three molecular descriptors of PFASs, molecular flexibility, atomic polarizabilities and the number of carbon atoms were the crucial characteristics to the binding with L-FABP. Further individual and joint effects analysis with BKMR model revealed that molecular flexibility could interact with other molecular properties to influence the affinity to L-FABP indirectly. As a parallel research content to BKMR, a machine learning based hepatotoxicity prediction model was constructed for efficient prediction of potentially hazardous PFASs. The current study could contribute to the more particular knowledge of hepatotoxicity of different PFASs, and the obtained toxicity prediction model could facilitate efforts in determining the toxicity of emerging PFASs and further assisting the management of toxic compounds.

CRediT authorship contribution statement

Jiayi Zhao: Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft; **Xiaoyue Shi**: Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft; **Zhiqin Wang**: Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft; **Sijie Xiong**: Methodology, Investigation, Data curation; **Yongfeng Lin**: Funding acquisition, Writing – review & editing; **Xiaoran Wei**: Visualization, Funding acquisition, Writing – review & editing, Supervision; **Yanwei Li**: Funding acquisition, Writing – review & editing; **Xiaowen Tang**: Resources, Conceptualization, Software, Funding acquisition, Project administration, Validation, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Supplementary materials

Structure of L-FABP complexed with two molecules of oleic acid (Fig. S1), Pearson correlation between binding affinity and dissociation constant (Fig. S2), top ten models of combinations with different chemical descriptors and machine learning methods (Fig. S3), performance of toxicity prediction models of PFASs towards portal of L-FABP (Fig. S4), univariate exposure-response functions for concerned PFASs descriptors (Fig. S5), descriptions for the used fingerprints (Table S1), hyper-parameters for algorithm optimization (Table S2), and code of the best machine learning model.

List of 1599 PFASs used in the present study (Supplementary materials_1), and detailed hyperparameter of the machine learning models (Supplementary materials_2).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2023.115310.

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